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Economics 103
Introduction to Econometrics
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PRACTICE Final Examination – WITH ANSWERS
March 2014

These is a three-hour open-book exam in which you are allowed to use your notebook, your textbook and a calculator.

Please answer ALL questions in all parts of the exam. Please choose the best answer among all available answers. Only one answer is the best answer! Please choose one, and only one, answer. If more than one answer is marked the question will not be counted.

You are provided with a results appendix labelled **STATA OUTPUTS**. Parts IV of the exam refers to this appendix.

Please type your name and your UCLA ID:

First Name	
Last Name	
UCLA ID #	

Please start solving the examinations only when you are instructed to do so.
Please stop immediately when instructed to do so.

Good Luck!

Part I

Part I—Theoretical Questions (each question worth 2 points):
Consider the following regression model:

$$y_i = \beta_1 + \beta_2 x_{2i} + \cdots + \beta_K x_{Ki} + e_i.$$

1. The interpretation of the slope coefficient β_k in the model is as follows:

   (a) a 1% change in x_{ki} is associated with a β_k% change in y_i.

   (b) a 1% change in x_{ki} is associated with a β_k% change in y_i, holding all other $K-1$ regressors constant.

   (c) a change in x_{ki} by one unit is associated with a β_k change in Y.

   (d) a change in x_{ki} by one unit is associated with a β_k change in Y, holding all other $K-1$ regressors constant.

   ANSWER: D
   This follows from the defintion of the regression coefficients in multiple regression. When we interpret the coefficient of a specific independent variable in a multiple regression, we know that the other included independent variables are held constant.

2. Suppose that we run a regression with K variables and obtain the SSE for that regression, say SSE_K. Suppose now that we add an additional regressor, resulting in a regression with $K+1$ variables and obtain the SSE for that regression, say SSE_{K+1}.Then

   (a) $SSE_K \geq SSE_{K+1}$ always.

   (b) if adjusted $R^2 = R^2$ then $SSE_K = SSE_{K+1}$.

   (c) $SSE_K \leq SSE_{K+1}$ always.

   (d) We cannot determine the relationship between SSE_K and SSE_{K+1} based on the information provided.

   ANSWER: A
   When we add a new regressor to the regression, the R^2 always either increases or remains constant. Please also refer to the formula of R^2.

3. R^2 is a valid measure of the goodness of fit of the regression if

   (a) R^2 is the same as the adjusted R^2.

   (b) the regression has a constant term.

   (c) the regression has more than two variables, i.e., $K > 2$.

   (d) R^2 is always a valid measure of the goodness of fit.

   ANSWER: B
   The R^2 is defined as SSR/SST, and hence is in principle valid no matter what the regression specification. However, without a constant term $SST \neq SSR + SSE$. In that case, hence R^2 does not measure the fraction of the total variance in the outcome variable explained by the regression, and is not constrained between 0 and 1. Hence, it is not a sensible measure of the goodness of fit

4. In a given data set, the larger is R^2

   (a) the lower is SSE.

   (b) the larger is SSE.

   (c) the lower is SSR.

   ANSWER: A
   Refer to the formula of $R^2 = 1 - SSE/SST$. Note that SST is uniquely pinned down by the data set.

5. Which of the following statements is not true:

   (a) a high R^2 or $\bar{R}^2$ does not mean that the regressors are a true cause of the dependent variable.

   (b) a high R^2 or $\bar{R}^2$ does not mean that there is no omitted variable bias.

   (c) a high R^2 or $\bar{R}^2$ always means that an added variable is statistically significant.

   (d) a high R^2 or $\bar{R}^2$ does not necessarily mean that you have the most appropriate set of regressors.

   ANSWER: C
   R^2 does not indicate anything regarding any single regressor, since it is a measurement of the joint goodness of fit. It also does not say anything more about an added variable beyond that some additional variance of the outcome variable may be explained by inclusion of that variable.

6. Choosing the right functional form for the regression is important because

   (a) it allows one to capture the specific features of the observed data.

   (b) it assists one in collecting the right data for the empirical analysis.

   (c) it permits one to use the R^2 as a measure for the model specification.

   (d) it prevents one from including irrelevant variables in the regression.

   ANSWER: A
   In class, we have referred to the question of whether a relationship between the outcome variable and a regressor is linear or nonlinear as the question regarding functional form. Hence a) is the correct option. Note that in a broad sense "functional form" is sometimes also referred to the appropriate set of regressors to be included in the regression model. However, this is not how the concept was used in class.

7. If you had a two regressor regression model, then omitting one variable which is relevant

   (a) will have no effect on the coefficient of the included variable if the correlation between the excluded and the included variable is negative.

   (b) will always bias the coefficient of the included variable upwards.

   (c) can result in a negative value for the coefficient of the included variable, even though the coefficient will have a significant positive effect on Y if the omitted variable were included.

   (d) makes the sum of the product between the included variable and the fitted residuals different from 0.

ANSWER: C
Option(a) is wrong since the existence of bias has nothing to do with the sign of the correlation between the endogenous and exogenous vairables, it only determined by whether these two variables are correlated with each other. Option(b) is wrong since the bias can be both upward or downward. Option(d) is wrong since the sum of the product is always zero, independent from the regression assumptions. To see why the statement in option (c) is correct, refer to the formula of the omitted variable bias we discussed in lab lecture or the formula in the book. A pattern discussed in (c) could arise if either the effect of the omitted variable on the outcome is positive and the correlation of the omitted variable with the included regressor negative, or vice versa, the effect of the omitted variable on the outcome is negative and the correlation of the omitted variable with the included regressor positive.

8. As the sample size, say N, increases, with varying data points, the length of the $100\,(1-\alpha)$ confidence interval

   (a) increases.

   (b) decreases.

   (c) does not change.

   (d) increases in some cases and decreases in others.

   ANSWER: B
   If the sample size increases with varying data points (meaning that the additional values for x are different than the mean and hence add information that can be used to estimate the regression coefficients), then the standard error of parameters decreases. But when the standard error decreases, the length of the confidence interval decreases. Please also refer to the formulas of the estimated variance of OLS coefficients and interval estimation.

9. Which of the following will change if you scale the dependent variable in a simple regression model?

   (a) p-value

   (b) t-value of β_2

   (c) R^2

   (d) β_1

   ANSWER: D
   Rescaling of the dependent variable causes the changes of all the coefficients on the righthand side.

10. Which of the following is **NOT** an assumption of the Simple Linear Regression Model?

    (a) The value of y, for each value of x, is $y = \beta_1 + \beta_2 x + e$.

    (b) The variance of the random error e is $Var\,(e) = \sigma^2$.

    (c) The covariance between any pair of random errors e_i and e_j is zero.

    (d) The parameter estimate of β_2 is unbiased.

ANSWER: D
Please refer to the SR assumptions. Option(d) is the result of the SR assumptions, but it is not an assumption itself.

11. When collinear variables are included in an econometric model, coefficient estimates are

    (a) biased downward and have smaller standard errors.

    (b) biased upward and have larger standard errors.

    (c) biased and the bias can be negative or positive.

    (d) unbiased but they have larger standard errors.

    ANSWER: D
    Unbiasedness is only related to the MR assumptions. As long as the expectation of the error term is zero and the regressors are nonrandom, then all the coefficient estimates are unbiased. However, the collinearity problem can cause the standard error of estimators to increase. Please also refer to the formula of the estimated variance of estimators in the MR model when K=3.

12. If one rejects the null hypothesis that H_0: $\beta_k = 0$ against H_1: $\beta_k \neq 0$ at the significant level α, then

    (a) he/she will not reject it for H_0: $\beta_k = 0$ against H_1: $\beta_k > 0$.

    (b) It cannot be determined whether he/she will reject it for H_0: $\beta_k = 0$ against H_1: $\beta_k > 0$.

    (c) he/she will reject it for H_0: $\beta_k = 0$ against H_1: $\beta_k > 0$.

    (d) he/she will reject it for H_0: $\beta_k = 0$ against H_1: $\beta_k > 0$ but will not reject it for H_0: $\beta_k = 0$ against H_1: $\beta_k < 0$.

    ANSWER: B
    To answer this question, draw the rejection region of the two tests. You will find they are not completely overlapping.

13. One will reject the null hypothesis H_0: $\beta_k = 0$ against H_1: $\beta_k > 0$, at the significant level α

    (a) if and only if zeros does not belong to the $100\,(1-\alpha)$ confidence interval for β_k.

    (b) if and only if zeros belongs to the $100\,(1-\alpha)$ confidence interval for β_k.

    (c) the $100\,(1-\alpha)$ confidence interval cannot help in testing this hypothesis.

    (d) There is no relation between confidence interval and hypothesis testing.

    ANSWER: C
    The $100(1-\alpha)$ *confidence interval is related to the testing result of the two sided test (*$H_0 : \beta = 0$, $H_1 : \beta \neq 0$),*not to the one-tailed test stated in the question.*

14. To decide whether $Y_i = \beta_0 + \beta_1 X_i + u_i$ or $ln(Y_i) = \beta_0 + \beta_1 X_i + u_i$ fits the data better, you cannot consult the regression R^2 because

    (a) ln(Y) may be negative for $0 < Y < 1$.

    (b) the SST are not measured in the same units between the two models.

(c) the slope no longer indicates the effect of a unit change of X on Y in the log-linear model.

(d) the regression R^2 can be greater than one in the second model.

ANSWER: B
You can not compare the R^2 of two regression models when the dependent variables are not measured in the same unit, because the SST is not the same. Hence, the R^2 do not refer to the fraction variance explained of the same SST.

15. To test whether or not the population regression function is linear rather than a polynomial of order r,

(a) check whether the regression R^2 for the polynomial regression is higher than that of the linear regression.

(b) compare the SST from both regressions.

(c) look at the pattern of the coefficients: if they change from positive to negative to positive, etc., then the polynomial regression should be used.

(d) use the test of (r-1) restrictions using the F-statistic.

ANSWER: D
This question explicitly calls for a test. Hence, while the relevant F-test statistic can be calculated using R^2 or the SST, a simple comparison of these statistics is not sufficient, a test is required.

16. The binary variable interaction regression

(a) can only be applied when there are two binary variables, but not three or more.

(b) is the same as testing for differences in means.

(c) cannot be used with logarithmic regression functions because it is not defined.

(d) allows the effect of changing one of the binary independent variables to depend on the value of the other binary variable.

ANSWER: D
Please refer to the definition of binary variables. Also compare the marginal effects of a binary variable when there is an interaction term and when there is not.

17. Consider the following multiple regression models (a) to (d) below. $DFem = 1$ if the individual is a female, and is zero otherwise; DMale is a binary variable which takes on the value one if the individual is male, and is zero otherwise; $DMarried$ is a binary variable which is unity for married individuals and is zero otherwise, and $DSingle$ is $(1 - DMarried)$. Regressing weekly earnings ($Earn$) on a set of explanatory variables, you will experience perfect multicollinearity in the following cases except:

(a) $Earn_i = \beta_0 + \beta_1 DFem_i + \beta_2 DMale_i + \beta_3 X_{3i}$

(b) $Earn_i = \beta_0 + \beta_1 DMarried_i + \beta_2 DSingle_i + \beta_3 X_{3i}$

(c) $Earn_i = \beta_0 + \beta_1 DFem_i + \beta_3 X_{3i}$

(d) $Earn_i = \beta_0 DFem_i + \beta_1 Male_i + \beta_2 DMarried_i + \beta_3 DSingle_i + \beta_5 X_{5i}$

ANSWER: C
Please refer to the definition of perfect collnearality. The typical case when perfect collinearity arises is when indicator variables for all values of a categorical values are included, since then these indicator variables sum to 1, which is equal to the constant term.

18. In the regression model $Y_i = \beta_0 + \beta_1 X_i + \beta_2 D_i + \beta_3 (X_i D_i) + u_i$, where X_i is a continuous variable and D_i is a binary variable, to test that the effect of X_i on Y_i is identical for both values of D_i, you must use

    (a) a separate t-test for $H_0 : \beta_2 = 0$ and $H_0 : \beta_3 = 0$.

    (b) F-test for the joint hypothesis that $\beta_0 = 0$, $\beta_1 = 0$.

    (c) a t-test for $H_0 : \beta_3 = 0$.

    (d) F-test for the joint hypothesis that $\beta_2 = 0$, $\beta_3 = 0$.

    ANSWER: C
    The coefficient of the interaction term measures the difference of the effect of two groups. If it is zero, then there is no different effect.

19. A large company is accused of gender discrimination in wages. The following model has been estimated from the company's human resource information

    $$\widehat{\ln(Wage)} = 1.439 + 0.0834(Edu) + 0.0512(Exper) + 0.1932(Male)$$

    where WAGE is hourly wage, Edu is years of education, $Exper$ is years of relevant experience, and $Male$ indicates the employee is male. What hypothesis would you test to determine if the discrimination claim is valid?

    (a) $H_0 : \beta_{Male} = 0$; $H_1 : \beta_{Male} \geq 0$

    (b) $H_0 : \beta_{Male} = \beta_{Edu} = \beta_{Exper} = 0$; $H_1 : \beta_{Male} \neq 0$ *and* $\beta_{Exper} \neq 0$ *and* $\beta_{Exper} \neq 0$

    (c) $H_0 : \beta_{Male} = \beta_{Edu} = \beta_{Exper} = 0$; $H_1 : \beta_{Male} \neq 0$ *or* $\beta_{Exper} \neq 0$ *or* $\beta_{Exper} \neq 0$

    (d) $H_0 : \beta_{Male} \leq \beta_{Edu} : or : \beta_{Male} \leq \beta_{Exper}; : H_1 : \beta_{Male} > \beta_{Edu} : or : \beta_{Male} > \beta_{Exper}$

    ANSWER: A
    β_{MALE} captures the difference in mean log wages of males and females, and hence is a a measure of discrimination in wages. Since the regression controls for years of education and years of experience, wage differences between gender due to differences in education and experience are controlled for. Hence, the coefficient on MALE mean difference in wages holding education and experience constant.

20. When an exogenous instrument is used, IV estimators are

    (a) consistent and approximately normally distributed in large samples

    (b) unbiased and BLUE in all sample sizes

    (c) consistent if z is normally distributed

    (d) normally distributed in all sample sizes and consistent in large sample sizes

ANSWER: A
Please refer to the definition and the computation of the instrumental variables (IV) estimator. The whole point of this approach is that the IV estimator is consistent (even though the OLS estimator might not be). In addition, it turns out to be approximately normally distributed in large samples as well.

21. Which of the following statements is true regarding $var(\hat{\beta}_2)$ when estimated by IV using z as an instrument for x?

    (a) instrumental variable estimation leads to larger variance of estimates compared to OLS
    (b) instrumental variable estimation leads to smaller variance estimates compared to OLS
    (c) the variance of the IV estimates can be larger or smaller than OLS, depending on r_{xz}^2
    (d) there is no way to know which estimator leads to estimates with a larger variance, it depends on the data

    ANSWER: A
    Please compare the formula of the estimated variance of the instrument variable estimators and the OLS estimators. The whole point of the IV estimator is to only use a part of the variation in the explanatory variable (that part which is not correlated with the error term). This is clear from the fact that in the second stage the predicted value for the explanatory variable is used. The variance of the predicted explanatory variable is smaller than the variance of the actual explanatory variable. Hence the variance is higher.

22. What is the null hypothesis when performing an F-test to test the strength of multiple instruments $j = 1, ..., J$?

    (a) the instruments are weak, with no coefficient θ_j on the instruments in the first stage different from 0.
    (b) the instruments are strong with all coefficients θ_j on the instruments significantly different from 0.
    (c) the instruments are sufficiently strong with a coefficient θ_j on at least one instrument significantly different from 0.
    (d) the instruments are weak with all coefficients θ_j on the instruments different from 0.

    ANSWER: A
    Please refer to the definition of the F-test and the first-step of the 2SLS estimation.

23. When you are implementing an instrumental variable regression, you are worried about

    (a) a potential direct effect of the instrumental variable on the outcome
    (b) a weak relationship between the instrumental variable and the endogenous variable
    (c) a remaining correlation of the instrumental variable and the error term
    (d) all of the above

    ANSWER: D
    A valid IV should be truly exogenous (no correlation with the error term), correlated with the endogenous variable, and should only affect the dependent variable through the endogenous variable (i.e., have no direct effect on the outcome).

Part II—Practical and Computational Questions (each question worth 2 points):

Consider the following regression model:

$$y_i = \beta_1 + \beta_2 x_{2i} + \cdots + \beta_K x_{Ki} + e_i,$$

24. You have estimated the following simple regression model

$$y = 379 + 1.44x^3 \tag{1}$$

What is the elasticity when x = 8.49?

(a) 263.19

(b) 311.39

(c) 2.10

(d) -24.7

ANSWER: C
$Elasticity = \frac{dy}{dx}\frac{x}{y}$
$\frac{dy}{dx} = 1.44 * 3 * x^2 = 4.32x^2$
$x = 8.49 \implies y = 379 + 1.44(8.49)^3 = 1260.22$
$\implies Elasticity = 4.32(8.49)^2 \frac{8.49}{1260.22} = 2.10$

25. You have estimated a two variable model, i.e., $K = 2$, and your printout includes the following information

$$\begin{aligned} s_{xy} &= 3614.00 \\ s_x &= 12.72 \\ s_y &= 394.61 \\ SST &= 758,912 \end{aligned}$$

Then the R^2 for this regression model is:

(a) .72

(b) .11

(c) .03

(d) .5

ANSWER: D
In the simple regression model, $R^2 = r_{xy}^2 = \left(\frac{s_{xy}}{s_x s_y}\right)^2$. *For the numbers of the question, we get that* $r_{xy} = 0.72$, *such that* $r_{xy}^2 = 0.52$. *Hence, (d) is the correct answer. Note that in the final either exact values will be given, or it will be stated explicitly that the result is approximate.*

26. Suppose that $K = 3$, $\overline{y} = 9$, $\overline{x}_2 = 3$, $b_2 = 1.2$, $\overline{x}_3 = 2$, $b_3 = 1.3$, then the estimate for β_1, b_1 is

(a) 2.8

(b) 3.8

(c) 8.3

(d) 2.5

ANSWER: A
For any regression model we know that evaluating the estimated equation at the means will return the mean.
Mathematically, $\overline{y} = b_1 + b_2\overline{x}_2 + b_3\overline{x}_3$
$\implies 9 = b_1 + 1.2(3) + 1.3(2) \implies b_1 = 2.8$

27. A company is accused of racial discrimination in wages. The following model has been estimated from the company's human resource information

$$\ln(WAGE) = 1.439 + .0834 \times EDU + .0512 \times EXPER + .1932 \times WHITE$$

where $WAGE$ is hourly wage, EDU is years of education, $EXPER$ is years of relevant experience, and $WHITE$ indicates that the employee is white. How much more do white employees at the firm earn, on average?

(a) \$1.21 per hour more than non-whites.

(b) \$19.32 per hour.

(c) 19.32% more than non-whites.

(d) \$19,320 more per year than non-whites.

ANSWER: C
The log-linear regression equation can be interpreted such that a one unit increase in an x_k *variable (independent variable) leads to approximately a* $(100)\beta_k\%$ *change in the y (dependent) variable.*

28. Randomized, controlled experiments are needed to accurately measure treatment effects without

(a) the expense of having to treat everyone.

(b) raising public debate.

(c) exposing everyone to untested treatments.

(d) omitted-variable bias.

ANSWER: D
One of the main purposes of a randomized, controlled experiment is to obtain estimates of the causal effect of a treatment that is not affected by omitted-variable bias.

29. The following economic model predicts whether a voter will vote for an incumbent school board member

$$INCUMBENT = \beta_1 + \beta_2 MALE + \beta_3 PARTY + \beta_4 MARRIED + \beta_5 KIDS,$$

where

$INCUMBENT = 1$ if the voter votes for them, 0 otherwise. $MALE = 1$ if the voter is a male. $PARTY = 1$ if the voter is registered with the same political party as the incumbent. $MARRIED = 1$ for married voters, 0 otherwise. $KIDS$ is the number of school age children living in the voter's house.

If you believe marriage affects male and female voters differently, which variable should you add to the economic model to allow you to test the hypothesis?

(a) $MALE \times PARTY$

(b) $MALE \times MARRIED$

(c) $MARRIED \times KIDS$

(d) $MARRIED \times PARTY$

ANSWER: B
Inclusion of the interaction $MALE \times MARRIED$ allows marriage to have different effects on men and women in terms of their likelihood of voting for an incumbent.

30. Consider the same setup as in the previous question. Suppose it is claimed that a married man is less devoted to the incumbent than a married woman. Then the results must have

(a) $\beta_2 < 0$

(b) $\beta_2 > 0$

(c) $\beta_2 + \beta_4 > 0$

(d) $\beta_2 + \beta_4 < 0$

ANSWER: A
In the setup in the previous question the variable marriage has the same effect on both men and women. Therefore, the claim simplifies to "a man is less devoted to the incumbent than a woman".

31. Consider the following regression model given by

$$y_i = 1.1 + 2x_{i2} + .5x_{i3} + e_i$$

with the variance-covariance matrix of the regression coefficients

$$\begin{pmatrix} .7 & .2 & .3 \\ .2 & .5 & -.2 \\ .3 & -.2 & .3 \end{pmatrix}.$$

The standard error for $2b_2 + b_3$ is

(a) 1.22

(b) 0.43

(c) 0.39

(d) 0.11

ANSWER: A
$var(2b_2 + b_3) = 4var(b_2) + var(b_3) + 4cov(b_2, b_3)$
$\implies var(2b_2 + b_3) = 4(.5) + .3 + 4(-.2) = 1.5$
$se(2b_2 + b_3) = \sqrt{var(2b_2 + b_3)} = \sqrt{1.5} = 1.22$

32. You have estimated the following equation:

$$TestScore_i = 607.3 + 3.85Income_i - 0.0423Income_i^2 \quad (2)$$

where TestScore is the average of the reading and math scores on the Stanford 9 standardized test administered to 5th grade students in 420 California school districts in 1998 and 1999. Income is the average annual per capita income in the school district, measured in thousands of 1998 dollars. The equation

(a) suggests a positive relationship between test scores and income until a value of the income variable of approximately 45.508.

(b) is positive until a value of Income of $610.81.

(c) does not make much sense since the square of income is entered.

(d) suggests a positive relationship between test scores and income for all of the sample.

ANSWER: A
To find the point at which a differentiable function changes slope, take the first derivative and set it equal to 0.
$\frac{dTestScore_i}{dIncome_i} = 3.85 - 0.0846Income_i^* = 0$
$\implies Income_i^* = 45.508$

Part III—Lab Questions (each question worth 1 point):

The questions in this part relate directly to the Lab class 103L.
Consider the following regression model:

$$y_i = \beta_1 + \beta_2 x_{2i} + \cdots + \beta_K x_{Ki} + e_i,$$

33. The essence of the regression analysis conducted by J.J. Espinoza is to

    (a) determine how Walt Disney Company should structure advertising in the course of the introduction of new movies.
    (b) evaluate the performance of the analysts at Walt Disney Company.
    (c) determine who to award financial aid to from a selected minority group.
    (d) evaluate the performance and labor market outcome of Hispanic students.

    ANSWER: A
    In his guest lecture, JJ described how he uses regression analysis to best time the advertisements of new movies.

34. The objective of the analysis performed by Vikas Gupta, for Factual, is to

    (a) determine the main characteristics of individuals who search for mortgages.
    (b) provide an evaluation those applying for access to special internet sites.
    (c) deliver the most relevant ads through mobile devices.
    (d) conduct market research for new electronic devices.

    ANSWER: C
    In his guest lecture, Vikas described how his company processes cell phone GPS data to construct tiers of individuals with similar characteristics to which his clients can deliver addvertisements to.

35. The approach of calculating the regression coefficients in three steps discussed in the lab lecture

    (a) is a way to get approximate values of the multiple regression coefficient estimates
    (b) is a way to get exact values of the multiple regression coefficient estimates
    (c) is a way to obtain estimates of the causal effect
    (d) can only be used if assumptions MR1-MR5 hold

    ANSWER: B
    The three-step method discussed in the lab lecture is equivalent to regular estimation of multiple regression, i.e., the exact identical coefficient estimates are obtained (this is based on the so-called Frisch-Waugh theorem).

36. Regressing average TESTSCR on the average student teacher ratio (STR) in school districts in California we had found

    (a) that larger classes have a positive causal effect on testscores, but the effect is moderate
    (b) that larger classes have a negative causal testscores, but the effect is moderate

(c) that larger classes are associated with smaller testscores, but the effect is moderate relative to the mean testscore

(d) that larger classes are associated with smaller testscores, but the size of the effect is hard to interpret

ANSWER: C
We found a negative coefficient of STR, but concluded that a) relative to the mean TESTSCR the effect is moderate and b) that this effect is likely to at least in part to be due to omitted factors.

37. The coefficient on STR in the regression of TESTSCR on STR

(a) is likely to yield the causal effect of class size on test scores

(b) is likely to yield a biased coefficient of the effect of class size on testscores

(c) cannot be interpreted since STR has a different scale than TESTSCR

(d) cannot be interpreted because the true relationship is not linear

ANSWER: B
We had discussed how other factors are likely to be present that determine TESTSCR and are correlated with STR.

38. Suppose we failed to include the variable SUBSLUNCH indicating the fraction of students receiving a subsidized lunch in the regression of TESTSCR on STR. Let β_3 be the coefficient on SUBSLUNCH were it to be included in the regression. The omitted variable bias is

(a) β_3

(b) $cov(STR, SUBSLUNCH)/var(STR)$

(c) $cov(TESTSCR, SUBSLUNCH)/var(STR)$

(d) $\beta_3 cov(STR, SUBSLUNCH)/var(STR)$

ANSWER: D
We had derived the omitted-variable bias formula based on the probability limit of b_2, and found that the bias (the difference between the probability limit of b_2 and the true β_2) is equal to $\beta_3 cov(STR, SUBSLUNCH)/var(STR)$. The book presents a very similar formula based on the expectation rather than the probability limit.

Part IV Demand for Meat (each question worth 2 points)

This question refers to the output labelled "The Demand for Meat" provided in the **STATA Output**. Note that the stata command "matrix list e(V)" is the same as "estat vce".

39. The F-statistic for H_0: $\beta_2 = \beta_3 = \beta_4 = 0$ from the first regression is (approximately):

(a) $25,717.38$

(b) 8.08

(c) 18.12

(d) 1.70

ANSWER: D
Simply observe the F-statistic reported in the output of the first regression. This is precisely the null hypothesis that is being tested.

40. The results indicate that

(a) there is high degree of collinearity between income and prices.

(b) the coefficient on the variable income is significant, because all the others are not significant.

(c) one should include in the regression the interactions between all prices.

(d) it is essential to have the variable income in the regression.

ANSWER: D
Observe that in the second regression, income is highly significant (very low p-value). Furthermore, the standard errors of the coefficients on prices are all substantially reduced, which is the opposite of what we would expect if there was a high degree of collinearity between income and prices. Instead, including income reduces the variance of the error of the regression (see the values for the root MSE), and hence reduces estimates of the standard errors.

41. The results of the first regression indicate that the hypothesis H_0: $\beta_2 = \beta_3 = \beta_4 = 0$

(a) is true for any $\alpha \leq .10$.

(b) H_1 is true for any $\alpha > .10$.

(c) neither H_1 nor H_0 are true.

(d) We cannot conclude from the results obtained which hypothesis is true.

ANSWER: A
The output for regression 1 displays the p-value on the F-statistic as 0.1921 so for any $\alpha < 0.1921$ *we can not reject the null hypothesis.*

42. The point estimates from the second regression indicate that

(a) Meat and vegetables and fruit are substitutes, while meat and bread and cereal are complements.

(b) Meat and vegetables and fruit are complements, as are meat and bread and cereal.

(c) Meat and vegetables and fruit are complements, while meat and bread and cereal are substitutes.

(d) Vegetables and fruit and bread and cereal are substitutes.

ANSWER: A
If x and y are complements, an increase in the price of x will lead to a decrease in the quantity demanded of y. If x and y are substitutes an increase in the price of x will lead to an increase in the quantity demanded of y. Look at the sign of the coefficients of the relevant variables to make an assessment.

43. Based on the results of the second regression, the t-statistic for the test of H_0: $\beta_3+\beta_4 = 0$ against H_1: $\beta_3 + \beta_4 \neq 0$ will be

(a) -0.6785

(b) 1.09

(c) -2.584

(d) -1.51

ANSWER: A

Quick hint: if $\hat{\lambda}$ is a random variable and you are testing $H_0 : \lambda = c$ vs. $H_1 : \lambda \neq c$ then $t - stat = \frac{\hat{\lambda}-c}{se(\hat{\lambda})}$

$se(\hat{\lambda})$ is always positive so the sign of $\hat{\lambda} - c$ determines the sign of the t-stat.

In this case $\lambda = \beta_3 + \beta_4$.

$se(\hat{\beta}_3 + \hat{\beta}_4) = \sqrt{var(\hat{\beta}_3) + var(\hat{\beta}_4) + 2cov(\hat{\beta}_3, \hat{\beta}_4)}$

Check the third stat output table to find the appropriate variances.

$\implies se(\hat{\beta}_3 + \hat{\beta}_4) = \sqrt{0.0214 + 0.0498 - 2(0.0011)} = 0.2627 \implies t - stat = \frac{0.1590-0.3372}{0.2627} = -.678$

44. Consider testing the restriction in model 2 that $\beta_5 = 0$

(a) this can be done based on a t-test

(b) this can be done using an F-test based on SSR in model 1 and model 2

(c) this can be done using an F-test based on R^2 in model 1 and model 2

(d) all of the above

ANSWER: D

The slides show that a t- and F-test are equivalent if we are testing only one restriction. Note that $SST = \sum (y_i - \bar{y})^2$ is the same in both models, since the restrictions under the null hypothesis do not lead to a modification of the dependent variable.

Note that model 1 is the restricted model, and model 2 is the unrestricted model in this case. Hence, the basic definition of the F-test statistic is $F = \frac{SSE_1-SSE2}{SSE2}\frac{N-K}{1}$.

Therefore, $R_1^2 = 1 - \frac{SSE_1}{SST}$ and $R_2^2 = 1 - \frac{SSE_2}{SST}$

$\implies R_2^2 - R_1^2 = \frac{SSE_1-SSE_2}{SST}$

and $1 - R_2^2 = \frac{SSE_2}{SST}$

$\implies \frac{R_2^2-R_1^2}{1-R_2^2} = \frac{SSE_1-SSE_2}{SSE_2}$

The F-stat can then be computed with the correct degrees of freedom and number of restrictions. Similalry, for the SSR we have that $SST = SSE+SSR$ and hence $SSE = SST-SSR$. Using the definition of the F-statistic, we have that $F = \frac{SSR_2-SSR1}{SST-SSR_2}\frac{N-K}{1}$. Note that deviding the numerator and denominator by SST gives the R^2 version of the test.

45. The large constant estimate in both regressions suggests that

(a) prices cannot affect the consumption of an essential good, such as meat.

(b) there is no implication regarding the impact of the other variables.

(c) there is almost no variability in the consumption of meat.

(d) the adjusted R^2 in the first regression should be smaller than R^2.

ANSWER: B
Note that the constant is the level of meat purchases predicted at a price of meat of 0. In this case, we are very unlikely to ever observe a price of zero or near zero, and hence the constant term cannot be interpreted directly. Instead, it sets the level of the prediction of meat purchases based on all the regressors in the model. More generally, the value of the constant doesn't change the interpretation of the signs (or magnitudes) of the slope coefficients.

46. The interval estimate for the coefficient on income, i.e., β_5, for $\alpha = .05$ is approximately

    (a) $[-1.566, +2.912]$
    (b) $[+1.566, +2.912]$
    (c) $[-0.142, +0.460]$
    (d) $[2 \times 0.142, 2 \times 0.460]$

    ANSWER: B
    Look at the 95% confidence interval in output table 2 on the coefficient for y.

47. The fact that R^2 in model 2 is substantially higher than in model 1 means:

    (a) that income explains an important part of the variance of meat purchases
    (b) that income has a causal effect on meat purchases
    (c) that income is exogenous
    (d) that the coefficients on the variables are biased

    ANSWER: A
    R^2 is a measure of how much variation in the dependent variable is explained by the model. Hence if R^2 goes up substantially when income is added to the model you can then conclude that income explains an important part of the variance of meat purchases.

48. The fact that coefficients on prices in model 2 barely change as income is included as regressor implies

    (a) that these coefficients measure the causal effect of these prices on meat purchases
    (b) that these coefficients do not measure the causal effect of these prices on meat purchases
    (c) that income is likely heterogeneous in the population
    (d) that there is a low correlation of individual income and these prices

    ANSWER: D
    We can't conclude that the coefficients measure the causal effect of these prices on meat purchases since there still is potential omitted-variable bias. If there were high correlation between individual income and these prices we would expect to an omitted-variable bias in the regression which does not include income. The sign and magnitude of the omitted-variable bias depends on $cov(p_i, income)$ for all prices, as well as the sign and size of the effect of income on meat purchases.

49. The elasticity of meat purchases with respect to a change in price of meat

    (a) cannot be calculated without further information

(b) is -67.63

(c) is -0.6756401

(d) is economically significant

ANSWER: A
We need to know the quantity of meat for which we will evaluate the elasticity, as well as a level of the price. One approach would be to take the mean of the quantity of meat purchased and the mean of the price. Neither the means nor specific values for quantities or prices are provided, hence the answer is a).

50. The 95% confidence interval for the effect of price of meat on meat purchases

(a) is approximately [-1.1629,-0.1885]

(b) changes with the scaling of income

(c) is unaffected by the scaling of meat purchases

(d) all of the above

ANSWER: A
Scaling income doesn't affect the other regressors. Scaling the dependent variable (meat purchases) will re-scale all the coefficients, and hence their confidence intervals.